

LabLink: On-demand GPU teaching labs accessible through the browser

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DOI: 10.xxxxxx/draft

Software

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Editor: Open Journals

Reviewers:

- @openjournals

Submitted: 01 January 1970

Published: unpublished

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Summary

Quantitative biology increasingly relies on GPU-accelerated deep learning tools with graphical interfaces, but teaching them hands-on is hard: installation consumes the opening hours of a workshop, and many participants' laptops have no GPU. LabLink is an open-source Infrastructure-as-Code platform with which a research group deploys reproducible GPU desktops in its own Amazon Web Services (AWS) account or on any machines that run Docker. Each participant claims a seat by entering an email address and lands in a full Linux desktop with the instructor's software preinstalled, in a browser tab (Figure 1). In our benchmark, a 30-seat pool was ready about seven minutes after a cold start. LabLink has served approximately 600 participants across 15 events.

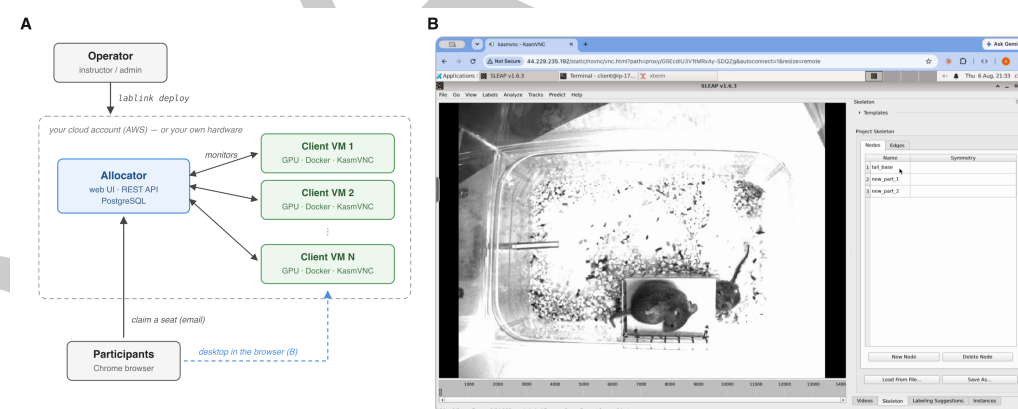


Figure 1: LabLink overview. (A) An operator deploys the allocator into their own cloud account or onto a lab-owned server with one CLI command; it provisions or registers one client machine per participant and monitors its health. Participants enter an email address and receive a Linux desktop streamed to their browser. (B) SLEAP v1.6.3 running in a LabLink browser desktop.

Statement of need

Hands-on workshops are how research software spreads (Wilson, 2016), but GPU-dependent software is hard to teach. Installation means resolving dependency stacks, GPU drivers, and operating-system-specific configuration (Mangul et al., 2019), and the first hours of a workshop are routinely lost to it. Many participants also have no local GPU, so even a successful installation cannot run training or inference.

The common workarounds each sacrifice something: preinstallation shifts the burden onto the

least experienced users, shared teaching servers require institutional infrastructure most labs lack, and cloud notebooks do not, by default, run graphical desktop applications. LabLink closes this gap: a lab creates a pool of identical, isolated, GPU-backed Linux desktops in its own cloud account or on machines it already has, hands each participant one link, and tears everything down when the workshop ends. It targets maintainers and instructors of GPU-dependent scientific software who must equip a room without per-laptop setup or institutional computing infrastructure.

State of the field

Domain platforms such as brainlife.io (Hayashi et al., 2024), Neurodesk (Renton et al., 2024), brainchop (Masoud et al., 2023), and NeuroCAAS (Abe et al., 2022) serve their own catalogs of neuroimaging environments or batch analyses, not software the instructor packages.

General-purpose teaching infrastructure is notebook-centric. JupyterHub (Project Jupyter contributors, 2026a) can stream a Linux desktop through jupyter-remote-desktop-proxy (Project Jupyter contributors, 2026b), but as a standing service someone must keep running: on one server with The Littlest JupyterHub (Project Jupyter contributors, 2026c), on a Kubernetes cluster for larger classes and for BinderHub (Project Jupyter et al., 2018), or hosted by a provider such as 2i2c (2i2c, 2026). LabLink starts one level down: from an image of the instructor's software and the lab's cloud credentials, one command provisions a GPU machine per participant and a second removes all of it, so the lab operates nothing between events. Google Colab (Google LLC, 2026b) offers GPUs only in notebooks, with session limits and no instructor control, and Galaxy's Training Infrastructure as a Service (Rasche et al., 2023) serves Galaxy workflows on shared public servers.

Among browser remote desktops, Apache Guacamole (The Apache Software Foundation, 2026) gateways a browser to existing machines, Open OnDemand (Hudak et al., 2018) serves desktops from an institutional HPC cluster, Kasm Workspaces (Kasm Technologies, 2026a) runs containerized desktops as an account-based service, Azure Lab Services (Microsoft, 2026) provisioned per-student VM pools but retires in June 2027, and Amazon WorkSpaces (Amazon Web Services, 2026b) streams managed GPU desktops bound to AWS.

LabLink's contribution is the integration around the desktop rather than the desktop itself: one GPU machine per participant, provisioned reproducibly in infrastructure the operator owns, running instructor-packaged software, with countless email-based seat claiming, health monitoring with automatic recovery, and teardown when the event ends.

Software design

LabLink has two components. A single **allocator** per deployment holds the cloud credentials that run OpenTofu (OpenTofu contributors, 2026), the database of machines and seats, the participant-facing web page, and the nginx (nginx contributors, 2026) proxy through which desktops are reached. A **client service** on every machine registers it, reports heartbeats, GPU health, and seat status, ships its logs, and rotates the desktop password when a seat is claimed. Because every report is a client-initiated REST call, a machine needs no public address or inbound rule, so machines LabLink did not provision, even behind an institutional firewall, can join the same pool. Except for direct LAN access, desktops are streamed over WebSockets through the allocator, the deployment's single participant-facing endpoint.

The allocator is a Flask (Pallets, 2026) service backed by PostgreSQL (The PostgreSQL Global Development Group, 2026), whose row-level locking lets a burst of simultaneous claims each take a different free machine. OpenTofu provisions cloud machines in the background; only a small state bucket and lock table persist between events. On AWS the allocator sweeps the pool once a minute and reboots any machine that fails to start, fails its GPU check,

or goes silent, warm to preserve an assigned session; after three failures its seat is released. Bring-your-own hosts are monitored but not rebooted.

Each client machine runs one Docker (Merkel, 2014) container built from the instructor's image, hosting both the KasmVNC (Kasm Technologies, 2026b) desktop and the client service. A whole machine per participant trades cost for isolation: one crashed session or saturated GPU affects only its own seat. At us-west-2 on-demand prices (Amazon Web Services, 2026a), a g4dn.xlarge seat costs about \$0.53 per hour, roughly \$2 per participant for a four-hour workshop. KasmVNC replaced Chrome Remote Desktop (Google LLC, 2026a), used at LabLink's first events, which required a Google account per participant and could not be version-pinned.

The allocator's own infrastructure is defined in a separate [template repository](#), deployed from a laptop with `lablink-cli` or through the template's GitHub Actions workflows. The manual provider registers, monitors, and proxies bring-your-own machines but leaves their creation and teardown to the operator, so only the AWS provider delivers the full provision-to-destroy lifecycle. Each package is tested and published to PyPI independently, with 90% coverage enforced and the allocator's OpenTofu tests run against AWS in CI; production images are built only from published versions. Documentation is at <https://lablink.talmolab.org/>.

Security

TLS is a configuration option, not a default; a bare-IP deployment runs over plain HTTP. Operator pages require operator credentials; client machines hold only per-machine secrets minted at registration and, on AWS, accept inbound connections only from the allocator and operator SSH; participants are administrators inside their desktop container but not on the host. The seat-claim page is deliberately unauthenticated: an email address labels a seat rather than authenticating a user, acceptable only because a deployment is short-lived with a fixed pool: a leaked link can exhaust seats but not provision machines, one address holds one seat, and scheduled destruction bounds the exposure. Claiming a seat rotates the desktop password and binds the browser to a signed session cookie, so participants cannot open one another's seats.

Configuration and adaptability

One YAML file, read by both deployment paths and the allocator, is the version-controllable record of a deployment, with no per-setting overrides. It states where machines come from; how the allocator reaches each desktop and participants reach the allocator (the AWS private network, a LAN, a WireGuard mesh overlay, or a client-initiated tunnel, plus Tailscale Funnel (Tailscale Inc., 2026) or Cloudflare Tunnel (Cloudflare, Inc., 2026) for an allocator without a public address); and a startup script that installs the instructor's software at boot, with retries because parallel installs fail intermittently at scale. Adapting to new software means editing these two files.

Benchmark

We deployed pools of 5, 10, 30, 60, and 150 g4dn.xlarge seats in us-west-2, one run per pool size (Figure 2); script, metadata, and raw data are in `paper/fig2-harness/`. Readiness is measured from the pool-launch command until the allocator marks a VM assignable, excluding the one-time allocator deploy.

From a cold start, a ready 30-seat pool takes about seven minutes. Per-client median readiness stayed near five minutes from 5 to 60 seats and about six and a half at 150, the size of our largest event, where all VMs were ready 11 minutes after launch; the infrastructure apply grew from 83 s to 249 s while boot time grew only modestly. All 255 VMs reached ready; ten (two at 10 seats, eight at 150) needed one automatic reboot after failing at the same point in the client-image pull, a transient registry failure rather than faulty machines.

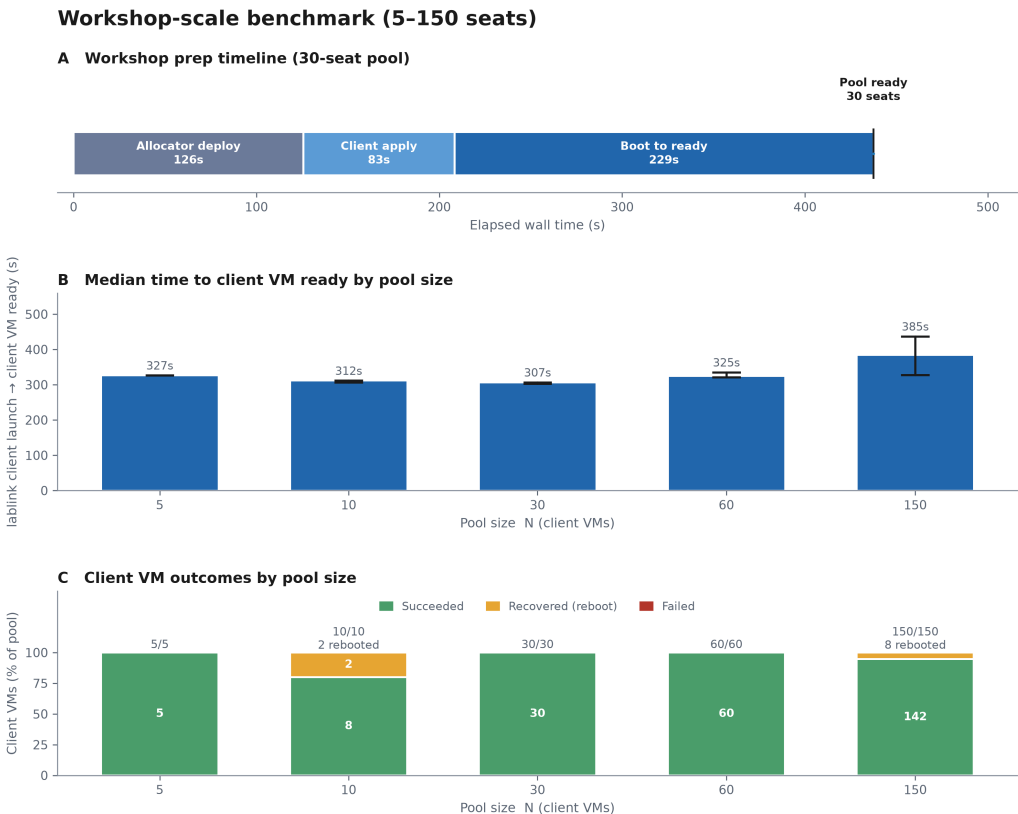


Figure 2: Workshop-scale benchmark, 5 to 150 seats. (A) Wall-clock to prepare one 30-seat workshop: allocator deploy (126 s), client-VM apply (83 s), and boot until all 30 VMs were ready (229 s). (B) Per-client readiness time (median and interquartile range) by pool size N. (C) Client VM outcomes per pool: ready directly, after one automatic reboot, or failed.

Research impact statement

LabLink was built to run SLEAP (Pereira et al., 2022) workshops and has delivered on-demand GPU desktops at 15 events between 2024 and 2026 (Table 1), from K–12 students to faculty, totaling approximately 600 attendances with nothing installed on participants' machines; the 2024 events ran on an earlier Chrome Remote Desktop-based implementation that predates the current repository.

LabLink's contribution to research is dissemination: installation is a documented barrier to adoption (Mangul et al., 2019), and a hands-on workshop is often a researcher's first chance to try GPU-dependent software. The approach generalizes beyond SLEAP: environments for Cellpose (Stringer et al., 2021), Kilosort4 (Pachitariu et al., 2024), idtracker.ai (Romero-Ferrero et al., 2019), and DeepLabCut (Mathis et al., 2018) each required changing only the configuration and startup script.

Table 1: Workshops and courses delivered on LabLink, 2024–2026. Participant counts marked ~ are organizer estimates.

Event	Date(s)	Participants	Location
Cosyne Main Tutorial – SLEAP	Feb 29, 2024	~150	Lisbon, Portugal
University of Illinois Urbana–Champaign workshop	Apr 2, 2024	~40	Urbana, IL, USA

Event	Date(s)	Participants	Location
Keiller Leadership Academy STEM Day	May 30, 2024	~125 (K–12 students)	San Diego, CA, USA
CAJAL Advanced Neuroscience Training Programme	Jun 2024	~20	Lisbon, Portugal
Salk EDGE Summer Program	Jul 24, 2024	3	La Jolla, CA, USA
NYU Langone Neuroscience T32 Workshop	Aug 19, 2024	~25	New York, NY, USA
Ellen Potter Research Connections	Apr 9–10, 2025	~20	La Jolla, CA, USA
Bridges to the Future Symposium	May 28, 2025	~50	La Jolla, CA, USA
CAJAL Advanced Neuroscience Training Programme	Jun 8–27, 2025	~30	Lisbon, Portugal
Zagreb NeuroData Summer School	Jul 2025	~30	Online / Zagreb, Croatia
Salk EDGE Summer Program	Jul 14–15, 2025	2	La Jolla, CA, USA
Computer Vision for Biologists	Sep 1–12, 2025	~30	Davis, CA, USA
Introduction to Multi-Animal Pose Tracking (US-RSE 2025)	Oct 6, 2025	~30	Philadelphia, PA, USA
Short course on machine learning applications	Oct 13–17, 2025	~30	Bar Harbor, ME, USA
CAJAL Advanced Neuroscience Training Programme	Jun 10, 2026	18	Lisbon, Portugal

AI usage disclosure

Claude Opus 5 and Claude Fable 5.1 (Anthropic), through Claude Code, assisted with both the software and this manuscript. The authors designed and wrote the architecture and initial implementation; Claude Code later drafted code and documentation, which the authors reviewed and validated with the test suites and in live deployments. The authors drafted the paper; Claude reviewed and shortened the text, checked citations, and helped assemble the benchmark figure. All text and figures were reviewed by the authors.

Acknowledgements

This work was supported by the NIH/BRAIN Initiative RF1 award under grant number 1RF1MH132653-01. We thank the open-source projects that made this work possible, including OpenTofu, Docker, KasmVNC, and PostgreSQL.

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